Problem M. MEX Problem

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	1024 megabytes

Little N has recently learned about MEX. MEX (Minimum Excluded) refers to the smallest non-negative integer that does not appear in an array. For example: $\text{MEX}([3,1,0]) = 2, \text{MEX}([2,2,1]) = 0, \text{MEX}([0,3,1,2]) = 4$. Little N defines that for a non-negative integer array of length k, $[b_1, b_2, \ldots, b_k]$, if $\text{MEX}([b_1, b_2, \ldots, b_k]) = k$, then the array b is called a k — good array.

Given a positive integer n, for an n — good array $p_1, p_2, \ldots, p_n$, Little N defines the function $f(p)$ as the number of integer pairs (l, r) that satisfy the following conditions:

- $1 \le l \le r \le n$;
- $p_l, \ldots, p_r$ is a $(r - l + 1)$ — good array.

Little N has proven the following theorem: for any n — good array p, there holds $1 \le f(p) \le n$. Now, Little N hopes you can find the number of n — good arrays p for each $i \in [1, n]$ such that $f(p) = i$. Of course, this number may be very large, so you need to provide the result modulo 998244353.

Input

The input consists of a single line containing an integer n ($1 \le n \le 10^5$).

Output

Output a single line containing n integers, where the i-th integer represents the number of n — good arrays p satisfying $f(p) = i$, modulo 998244353.

Examples

standard input	standard output
1	1
3	0 2 4
7	0 2312 1424 728 352 160 64

Note

For $n = 3$, it can be proven that there are six 3 — good arrays, with corresponding function values:

$$f([0,1,2]) = 3, f([0,2,1]) = 2, f([1,0,2]) = 3, f([1,2,0]) = 2, f([2,0,1]) = 3, f([2,1,0]) = 3$$